

# CBSE 2027 Physics — Predicted Paper (Set 3 of 10)

## SECTION A: Objective Type Questions (16 Marks)

*This section consists of 12 MCQs and 4 Assertion-Reasoning questions of 1 mark each.*

**Q1.** The electric potential at any point on the equatorial line of an electric dipole is: (a) maximum (b) zero (c) infinite (d) same as on the axial line (*Chapter: Electrostatic Potential and Capacitance*) | **[Confidence: Very High]**

**Q2.** Three identical capacitors, each of capacitance  $C$ , are connected first in series and then in parallel. The ratio of their equivalent capacitances  $C_s : C_p$  is: (a) **1 : 9** (b) **9 : 1** (c) **1 : 3** (d) **3 : 1** (*Chapter: Electrostatic Potential and Capacitance*) | **[Confidence: High]**

**Q3.** A potential difference  $V$  is applied across a copper wire of length  $L$ . If the length of the wire is doubled keeping the potential difference constant, the drift velocity of electrons will: (a) double (b) become zero (c) remain the same (d) be halved (*Chapter: Current Electricity*) | **[Confidence: Very High]**

**Q4.** A magnetic field  $B$  exists at the centre of a circular coil carrying current  $I$ . If the radius of the coil is halved and the current is doubled, the new magnetic field at the centre will be: (a)  $B$  (b)  $2B$  (c)  $4B$  (d)  $8B$  (*Chapter: Moving Charges and Magnetism*) | **[Confidence: High]**

**Q5.** A material suitable for making the core of a transformer should ideally have: (a) high retentivity and high coercivity (b) low retentivity and low coercivity (c) high permeability and low retentivity (d) low permeability and high coercivity (*Chapter: Magnetism and Matter*) | **[Confidence: Very High]**

**Q6.** A metal ring is held horizontally and a bar magnet is dropped through the ring with its length along the axis. The acceleration of the falling magnet is: (a) equal to  $g$  (b) less than  $g$  (c) greater than  $g$  (d) zero (*Chapter: Electromagnetic Induction*) | **[Confidence: Very High]**

**Q7.** In a series LCR circuit, at the resonant frequency, the power factor of the circuit is: (a) zero (b) 0.5 (c) 1 (d) infinite (*Chapter: Alternating Current*) | **[Confidence: Very High]**

**Q8.** If the electric field of an electromagnetic wave is oscillating along the Y-axis and its propagation is along the X-axis, the magnetic field must be oscillating along the: (a) X-axis (b) Y-axis (c) Z-axis (d) XY plane (*Chapter: Electromagnetic Waves*) | **[Confidence: High]**

**Q9.** Two thin lenses of power  $+5\text{ D}$  and  $-2\text{ D}$  are placed in contact coaxially. The focal length of the combination is: (a)  **$+33.3\text{ cm}$**  (b)  **$-33.3\text{ cm}$**  (c)  **$+30\text{ cm}$**  (d)  **$-30\text{ cm}$**  (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: Very High]**

**Q10.** In wave optics, the phase difference ( $\Delta\phi$ ) corresponding to a path difference of  $\lambda/4$  is: (a)  $\pi/4$  (b)  $\pi/2$  (c)  $\pi$  (d)  $2\pi$  (*Chapter: Wave Optics*) | **[Confidence: High]**

**Q11.** In a photoelectric effect experiment, the slope of the straight-line graph between stopping potential ( $V_0$ ) and frequency ( $\nu$ ) of incident light gives: (a)  $h$  (b)  $h/e$  (c)  $e/h$  (d) Work function (*Chapter: Dual Nature of Radiation & Matter*) | **[Confidence: Very High]**

**Q12.** The volume of a nucleus is directly proportional to its: (a) Mass number  $A$  (b) Atomic number  $Z$  (c)  $A^{1/3}$  (d)  $Z^{1/3}$  (Chapter: Nuclei) | [Confidence: Very High]

**Q13. Assertion (A):** The electric field is always directed perpendicular to an equipotential surface at every point. **Reason (R):** The work done in moving a test charge along an equipotential surface is zero. (Chapter: Electrostatic Potential and Capacitance) | [Confidence: Very High]

**Q14. Assertion (A):** An induced emf appears in any closed conducting coil in which the magnetic flux is changing over time. **Reason (R):** This is a direct consequence of Faraday's law of electromagnetic induction. (Chapter: Electromagnetic Induction) | [Confidence: Very High]

**Q15. Assertion (A):** Bohr's atomic model predicts that the energy of an electron in a hydrogen atom is quantized. **Reason (R):** Bohr postulated that the angular momentum of the orbiting electron is quantized as integral multiples of  $\hbar/2\pi$ . (Chapter: Atoms) | [Confidence: High]

**Q16. Assertion (A):** At absolute zero temperature (0 K), an intrinsic semiconductor behaves as a perfect insulator. **Reason (R):** At absolute zero, all valence electrons are tightly bound in covalent bonds and no free charge carriers are available for conduction. (Chapter: Semiconductor Electronics) | [Confidence: Very High]

---

## SECTION B: Very Short Answer Type Questions (10 Marks)

*This section consists of 5 questions of 2 marks each.*

**Q17.** The plot of terminal voltage ( $V$ ) versus current ( $I$ ) drawn from a cell is a straight line with a negative slope. Explain how you can determine the emf ( $\mathcal{E}$ ) and the internal resistance ( $r$ ) of the cell from this graph. (Chapter: Current Electricity) | [Confidence: High]

**Q18.** How is a moving coil galvanometer converted into a voltmeter of a specific range? Explain with the help of a circuit diagram and write the mathematical formula used. (Chapter: Moving Charges and Magnetism) | [Confidence: Very High]

**Q19.** Why are infrared waves commonly referred to as "heat waves"? Briefly state what vital role they play in maintaining the Earth's average temperature through the greenhouse effect. (Chapter: Electromagnetic Waves) | [Confidence: High]

**Q20.** Calculate the de-Broglie wavelength of an electron that has been accelerated from rest through a potential difference of 100 V. (Chapter: Dual Nature of Radiation and Matter) | [Confidence: Very High]

**Q21.** Draw a representative graph showing the variation of binding energy per nucleon ( $BE/A$ ) with the mass number ( $A$ ). On the graph, mark the regions where nuclear fusion and nuclear fission are most likely to occur. (Chapter: Nuclei) | [Confidence: Very High]

---

## SECTION C: Short Answer Type Questions (21 Marks)

*This section consists of 7 questions of 3 marks each.*

**Q22.** Derive an expression for the electric potential  $V$  at a distance  $r$  from an isolated point charge  $Q$ . Plot a graph showing the variation of electric potential  $V$  with distance  $r$ . (Chapter: *Electrostatic Potential and Capacitance*) | **[Confidence: High]**

**Q23.** A parallel plate capacitor is fully charged using a battery. After charging, the battery is disconnected, and a dielectric slab of dielectric constant  $K$  is fully inserted between the plates. Explain mathematically what happens to the capacitor's (i) capacitance, (ii) potential difference, and (iii) stored electrostatic energy. (Chapter: *Electrostatic Potential and Capacitance*) | **[Confidence: Very High]**

**Q24.** Two infinitely long straight parallel wires carry steady currents  $I_1$  and  $I_2$  in opposite directions. Derive the expression for the magnetic force per unit length acting between them. What is the nature of this force? (Chapter: *Moving Charges and Magnetism*) | **[Confidence: Very High]**

**Q25.** Define the term self-inductance. Derive an expression for the self-inductance of a long, air-cored solenoid of length  $l$ , cross-sectional area  $A$ , having a total of  $N$  turns. (Chapter: *Electromagnetic Induction*) | **[Confidence: Very High]**

**Q26.** Derive the formula  $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$  for refraction of a paraxial ray at a convex spherical surface separating two transparent media of refractive indices  $n_1$  and  $n_2$  ( $n_2 > n_1$ ). (Chapter: *Ray Optics and Optical Instruments*) | **[Confidence: Very High]**

**Q27.** Using Bohr's postulates for the hydrogen atom, derive the expression for the radius of the  $n^{\text{th}}$  stationary orbit of the electron. (Chapter: *Atoms*) | **[Confidence: Very High]**

**Q28.** Draw the V-I characteristics of a p-n junction diode in both forward bias and reverse bias. On the graph, clearly mark and define the terms 'threshold voltage' (cut-in voltage) and 'breakdown voltage'. (Chapter: *Semiconductor Electronics*) | **[Confidence: High]**

## SECTION D: Case-Study Based Questions (8 Marks)

*This section consists of 2 questions of 4 marks each. Questions usually have internal sub-parts.*

**Q29. Case Study 1: Young's Double Slit Experiment (YDSE)** *Passage:* In 1801, Thomas Young conclusively proved the wave nature of light through his double-slit experiment. In this setup, light from a single monochromatic source is passed through two narrow, closely spaced slits, acting as coherent sources. The superimposing light waves create an interference pattern of alternate bright and dark fringes on a distant screen. (a) What is meant by "coherent sources" of light? (1 mark) (b) Write the condition (in terms of path difference) for constructive interference to occur at a point on the screen. (1 mark) (c) If the entire YDSE apparatus is immersed in a liquid of refractive index  $\mu$ , what will happen to the fringe width? Justify your answer. **OR** If the ratio of the widths of the two slits is  $4 : 1$ , evaluate the ratio of maximum to minimum intensities ( $I_{\text{max}} : I_{\text{min}}$ ) in the interference pattern. (2 marks) (Chapter: *Wave Optics*) | **[Confidence: Very High]**

**Q30. Case Study 2: Transformers** *Passage:* The large-scale transmission of electrical energy over long distances is done with the use of transformers. A transformer is an electrical device used to step-up or step-down alternating voltage. It operates on a specific principle of electromagnetism and contains primary and secondary coils wound over a soft iron core. (a) State the underlying principle of a transformer. (1 mark) (b) Why is the soft iron core of a transformer laminated? (1 mark) (c) A step-down transformer reduces the supply voltage from **2200 V** to **220 V**. If the primary coil has **1000** turns, find the number of turns in the secondary coil. **OR** If the same transformer delivers an output power of **4.4 kW** at **220 V** with an efficiency of **80%**, calculate the current drawn by the primary coil. (2 marks) (*Chapter: Alternating Current*) | **[Confidence: Very High]**

## SECTION E: Long Answer Type Questions (15 Marks)

*This section consists of 3 questions of 5 marks each. All questions have an internal "OR" choice.*

**Q31. (a)** State Gauss's Law in electrostatics. **(b)** Using Gauss's Law, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius  $R$  and total charge  $Q$  at a point: (i) outside the shell ( $r > R$ ), and (ii) inside the shell ( $r < R$ ). **(c)** Plot a graph showing the variation of the electric field  $E$  with distance  $r$  from the centre of the shell. **OR (a)** Derive an expression for the electric field intensity at a point  $P$  situated at a distance  $r$  on the axial line of an electric dipole of dipole moment  $p$ . **(b)** An electric dipole of length **4 cm**, when placed with its axis making an angle of **60°** with a uniform electric field, experiences a torque of **4√3 Nm**. Calculate the magnitude of the electric field if the dipole has charges of **±8 nC**. (*Chapter: Electric Charges and Fields*) | **[Confidence: Very High]**

**Q32. (a)** Define the term 'drift velocity' of charge carriers in a conductor. **(b)** Deduce Ohm's Law ( $V = IR$ ) starting from the concept of drift velocity and the relaxation time of electrons. **(c)** Plot a graph showing the variation of electrical resistivity with temperature for (i) Copper (a metal), and (ii) Silicon (a semiconductor). **OR (a)** State the Biot-Savart Law expressing the magnetic field due to a current element in vector form. **(b)** Using Biot-Savart Law, derive the expression for the magnetic field at the centre of a circular coil of radius  $R$  carrying a current  $I$ . **(c)** State Ampere's Circuital Law and use it to find the magnetic field deep inside a long, tightly wound current-carrying solenoid. (*Chapter: Current Electricity / Moving Charges and Magnetism*) | **[Confidence: Very High]**

**Q33. (a)** Draw a labelled ray diagram of a Compound Microscope showing the image formation when the final image is formed at the least distance of distinct vision ( $D$ ). **(b)** Derive the expression for its total magnifying power ( $m$ ). **(c)** Why must both the objective and the eyepiece of a compound microscope have short focal lengths? Explain. **OR (a)** Draw a neat ray diagram showing the passage of a ray of light through a glass prism. **(b)** Derive the prism formula:  $\mu = \frac{\sin[(A+\delta_m)/2]}{\sin(A/2)}$ , where  $A$  is the angle of the prism and  $\delta_m$  is the angle of minimum deviation. **(c)** An equilateral glass prism has a refractive index of **1.5** in air. Calculate the angle of minimum deviation for this prism. ( $\sin 48.6^\circ \approx 0.75$ ) (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: Very High]**

---

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.